Lab Assignment: Non-preemptive Shortest Job First (SJF) CPU Scheduling Algorithm

Objective:

To understand and implement the Non-preemptive Shortest Job First (SJF) CPU scheduling algorithm and compute:
- Gantt Chart
- Completion Time (CT)
- Turnaround Time (TAT)
- Waiting Time (WT)
- Average Turnaround Time
- Average Waiting Time

Reference:

Step 1: Start

Step 2: Input

- Read number of processes n

Step 3: Read Process Details

For each process i = 1 to n:

- Read Arrival Time AT[i]
- Read Burst Time BT[i]
- Set Completed[i] = false

Step 4: Initialize

- current_time = 0
- completed_count = 0

Step 5: Repeat until all processes are completed

While completed_count < n do:

1. Set min_bt = ∞ // Initialize minimum burst time as very large
2. Set index = -1 // Initialize index to “no process selected”
3. For each process i = 1 to n:
    - If

§ AT[i] ≤ current_time // process has arrived

§ and Completed[i] = false //process not finished

§ and BT[i] < min_bt //has smaller burst time than currentminimum

then:

§ min_bt = BT[i] //update minimum burst time

§ index = I // store this process as candidate for execution

4. If index = -1:
   - No process has arrived
   - current_time = current_time + 1
5. Else:
   - current_time = current_time + BT[index]
   - CT[index] = current_time
   - TAT[index] = CT[index] - AT[index]
   - WT[index] = TAT[index] - BT[index]
   - Completed[index] = true
   - completed_count++

Step 6: Compute Averages

- Average Turnaround Time = ΣTAT / n
- Average Waiting Time = ΣWT / n

Step 7: Display Results

Step 8: Stop

```java
import java.util.*;

public class SJF_NonPreemptive {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter number of processes: ");
        int n = sc.nextInt();

        int[] AT = new int[n];
        int[] BT = new int[n];
        int[] CT = new int[n];
        int[] TAT = new int[n];
        int[] WT = new int[n];
        boolean[] completed = new boolean[n];

        for (int i = 0; i < n; i++) {
            System.out.print("Enter Arrival Time for P" + (i + 1) + ": ");
            AT[i] = sc.nextInt();

            System.out.print("Enter Burst Time for P" + (i + 1) + ": ");
            BT[i] = sc.nextInt();

            completed[i] = false;
        }

        int current_time = 0;
        int completed_count = 0;

        ArrayList<Integer> gantt = new ArrayList<>();
```

```java
        while (completed_count < n) {

            int min_bt = Integer.MAX_VALUE;
            int index = -1;

            for (int i = 0; i < n; i++) {
                if (AT[i] <= current_time && !completed[i] && BT[i] < min_bt) {
                    min_bt = BT[i];
                    index = i;
                }
            }

            if (index == -1) {
                current_time++;
            } else {
                gantt.add(index);

                current_time += BT[index];

                CT[index] = current_time;
                TAT[index] = CT[index] - AT[index];
                WT[index] = TAT[index] - BT[index];

                completed[index] = true;
                completed_count++;
            }
        }

        System.out.println(x: "\nProcess\tAT\tBT\tCT\tTAT\tWT");

        double totalTAT = 0, totalWT = 0;
```

```java
        System.out.println(x: "\nProcess\tAT\tBT\tCT\tTAT\tWT");

        double totalTAT = 0, totalWT = 0;

        for (int i = 0; i < n; i++) {
            System.out.println("P" + (i + 1) + "\t" + AT[i] + "\t" + BT[i] +
                    "\t" + CT[i] + "\t" + TAT[i] + "\t" + WT[i]);

            totalTAT += TAT[i];
            totalWT += WT[i];
        }

        System.out.println(x: "\nGantt Chart:");

        for (int i : gantt) {
            System.out.print("|  P" + (i + 1) + "  ");
        }
        System.out.println(x: "|");

        System.out.printf(format: "\nAverage Turnaround Time = %.2f", totalTAT / n);
        System.out.printf(format: "\nAverage Waiting Time = %.2f\n", totalWT / n);

        sc.close();
    }
}
```

```
Enter number of processes: 5
Enter Arrival Time for P1: 0
Enter Burst Time for P1: 7
Enter Arrival Time for P2: 2
Enter Burst Time for P2: 4
Enter Arrival Time for P3: 4
Enter Burst Time for P3: 1
Enter Arrival Time for P4: 5
Enter Burst Time for P4: 4
Enter Arrival Time for P5: 6
Enter Burst Time for P5: 6

Process AT      BT      CT      TAT     WT
P1      0       7       7       7       0
P2      2       4       12      10      6
P3      4       1       8       4       3
P4      5       4       16      11      7
P5      6       6       22      16      10

Gantt Chart:
|  P1  |  P3  |  P2  |  P4  |  P5  |
0       7       8       12      16      22

Average Turnaround Time = 9.60
Average Waiting Time = 5.20
```

```
Enter number of processes: 5
Enter Arrival Time for P1: 0
Enter Burst Time for P1: 6
Enter Arrival Time for P2: 1
Enter Burst Time for P2: 8
Enter Arrival Time for P3: 2
Enter Burst Time for P3: 6
Enter Arrival Time for P4: 3
Enter Burst Time for P4: 2
Enter Arrival Time for P5: 4
Enter Burst Time for P5: 4

Process AT      BT      CT      TAT     WT
P1      0       6       6       6       0
P2      1       8       26      25      17
P3      2       6       18      16      10
P4      3       2       8       5       3
P5      4       4       12      8       4

Gantt Chart:
|  P1  |  P4  |  P5  |  P3  |  P2  |
0       6       8       12      18      26

Average Turnaround Time = 12.00
Average Waiting Time = 6.80
```

```
Enter number of processes: 5
Enter Arrival Time for P1: 2
Enter Burst Time for P1: 5
Enter Arrival Time for P2: 0
Enter Burst Time for P2: 3
Enter Arrival Time for P3: 4
Enter Burst Time for P3: 5
Enter Arrival Time for P4: 1
Enter Burst Time for P4: 2
Enter Arrival Time for P5: 3
Enter Burst Time for P5: 4

Process AT      BT      CT      TAT     WT
P1      2       5       14      12      7
P2      0       3       3       3       0
P3      4       5       19      15      10
P4      1       2       5       4       2
P5      3       4       9       6       2

Gantt Chart:
|  P2  |  P4  |  P5  |  P1  |  P3  |
0       3       5       9       14      19

Average Turnaround Time = 8.00
Average Waiting Time = 4.20
```

Submitted By -: Vishal Verma(240691)